RESEARCH SUMMARY

TO ACCOMPANY STUDENT CHECKLIST (1A)

POSTPARTUM CUP

A LIFESAVING INNOVATION FOR REALTIME AND ACCURATE ESTIMATION OF OBSTETRIC BLOOD LOSS IN POSTPARTUM HAEMORRHAGE

RATIONALE OF THE STUDY :

BACKGROUND :

PPH (Post Partum Haemorrhage) is major cause of maternal death (~25-35%)

Globally one mother dies every 4 minute due to PPH.

India's MMR (Maternal Mortality Rate) is130. It is better than some of its neighbours, such as Pakistan (178), Bangladesh (176), Nepal (258). But it is substantially behind Sri Lanka (30), according to 2015 World Bank data. Fact Checker reported on June 19, 2018, India accounts for 17% of the global burden of maternal deaths. The leading cause of such deaths in India is PPH (38%). India's current MMR is still higher than the sustainable development goal (SDG) target- a set of globally agreed goals that India has signed on to is of 70 deaths per 1,00,000 live births for the world by 2030.

STATEMENT OF NECESSITY :

Every parturient woman is at risk of having PPH.

Severe PPH can lead to poor health of mother (maternal morbidity), and sometimes even death (mortality), particularly in low and middle-income countries. If the patient develops PPH, the golden hour (first 60 minutes) is the time at which resuscitation must be started to ensure best chances of survival. The probability of survival decreases sharply after the first hour, if resuscitation measures are not initiated in time. Early diagnosis and prompt treatment improve health outcomes for the mother. In many instances, the birth attendant assesses blood loss by looking at the amount of blood lost, and estimating its volume (visual estimation). While this method is not very accurate, it is available in all birth settings.

PPH is Easy to miss

Estimation of accurate blood loss in the present clinical scenario is very difficult and impossible with the prevalent and suggested methods like.

- A pictorial reference guide to aid visual estimation of blood loss.
- Blood drained into a fixed container for measurement.
- BRASSS-V DRAPE.
- Kelly's pad.

To curb the maternal mortality, early diagnosis even before appearance of clinical signs and symptoms is necessary. So that golden hour is not lost.

Ironically there is no ideal method to measure the post delivery blood loss.

Therefore, it is important to find the best method to measure blood loss after birth; one that is practical in all birth settings, including those in low to middle-income countries.

INNOVATION :

Inspiration :

As a growing child I have witnessed and experienced the long lasting emotional impact of maternal death during childbirth not only to the families who lost the life but also on the medical personnel treating the patient.

It inspired me for innovation of postpartum cup (PPH Cup), it is a silicone device.

Safety :

The certificate of analysis, (REPORT NO. 305 T-20 dated 02/10/20) from INDIAN DRUGS RESEARCH ASSOCIATION AND LABORATORY, PUNE- 411 005 (INDIA), tested it as non toxic and safe.

Social impact of the study :

The PPH Cup will be a one stop remedial solution for medical team to measure the blood loss after delivery in real-time and accurately.

It will be helpful in the low resource and primary health care setting in developing as well as developed countries. It will help in early diagnosis and prompt treatment thereby reducing the maternal mortality rate (MMR) and maternal morbidity. It will help the developing world to achieve the sustainable millennium development goal (SDG) of reduction in MMR. It will reduce the sanitary bio medical waste, so eco friendly. It is economical as well.

Objective of the study :

1) To know the safety and efficacy of the postpartum cup in measurement of the post delivery blood loss accurately and in real time.
2) To know the user friendliness and acceptance of patients, paramedics and obstetricians to the new device -PPH cup.
3) To know whether there is reduction in generation of the sanitary biomedical waste.
4) To get expert remarks and suggestions if any for future improvement of the product.

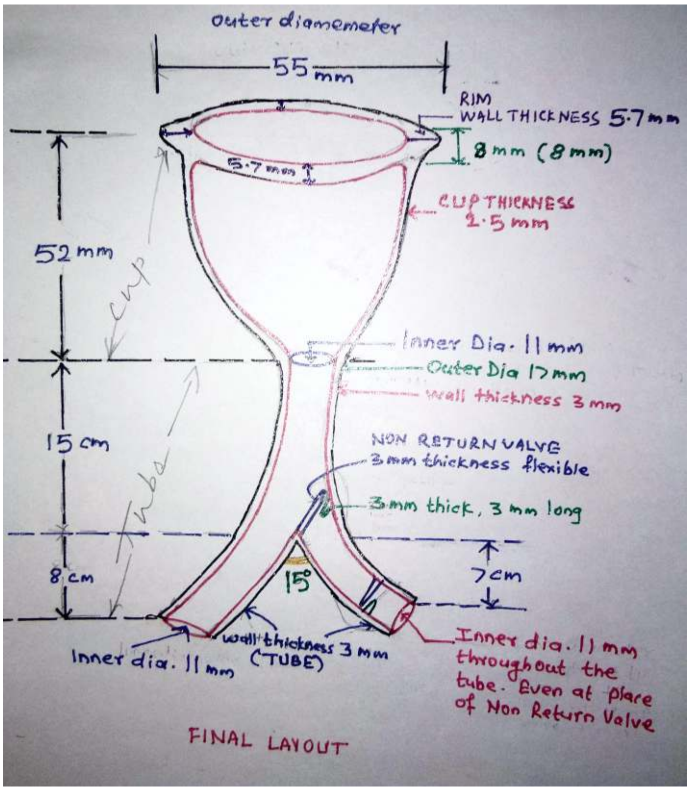
Design of study

It is prospective study involving only study group. Data was collected in prescribed survey proforma related to the participants (patients), paramedics and obstetrician. 600 participants were enrolled in the study.

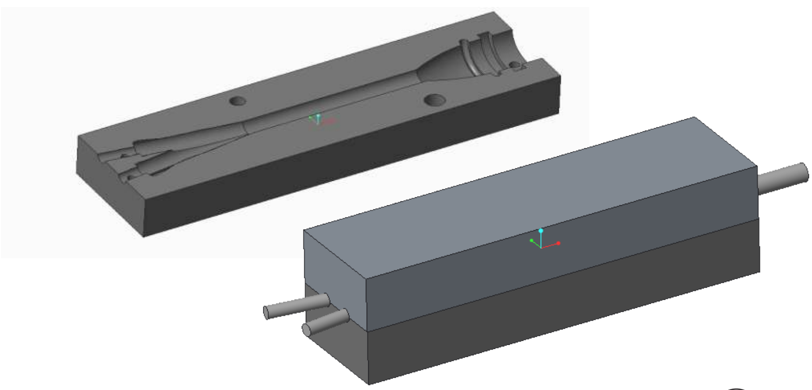
Project Design :

1) Preparing prototype

2) Design of PPH Cup



3) Developing Mould



Actual Mould

4) Actual PPH Cup

PPH Cup made up of silicone

5) Testing

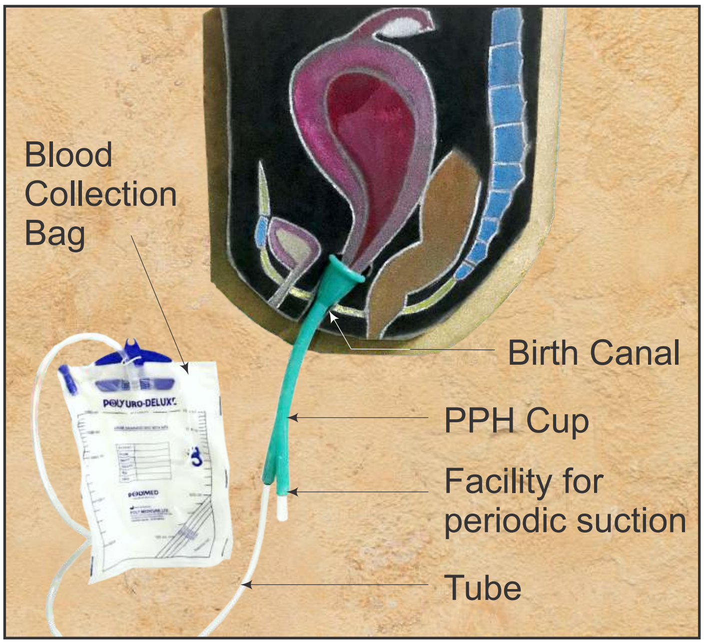
The testing was done for safety and toxicity from Indian Drug Research Laboratory, Pune.

Indian Drugs Research Association

6) Approvals

- **The approval for trial was obtained from the ethical committee of Medical College.**
- **Prior consent obtained from patients.**
- **Pilot study under the supervision of obstetricians.**

7) Working model of PPH Cup



8) Clinical Use

Place of study : Government Medical College, AKOLA-444001, Maharashtra State (INDIA).

Ethical Aspects :

The study was conducted after obtaining ethical clearance certificate from Institutional Review Board, Ethical Committee (IRB) Govt Medical College & Hospital, Akola. as per

- Ethical Guidelines for Biomedical Research on human subjects, Indian Council of Medical Research, New Delhi, 2006.
- National Guidelines for Ethical Committees reviewing biomedical and health research during COVID-19 pandemic were observed
- No information was collected related to the identity of the study participants.

Ethical Clearance Certificate :

From, institutional review board, Ethical committee,

Govt. Medical College & Hospital, Akola- 444001, Maharashtra State (INDIA).

(Out. no. GMCA/E.C./270/2020 Date : 09/03/2020)

Participants :

The participants in the research were all pregnant women (aged ≥18 years) admitted for childbirth.

Recruitment :

The participants admitted for childbirth with any type of delivery at place of study and willing to participate in the study after human informed consent were enrolled in the study.

Methodology of the study :

The research plan was submitted to the ethical committee (IRB) of Government Medical College, AKOLA - 444 001, Maharashtra State (INDIA) along with safety and toxicity reports and the PPH cups were supplied to the institution for intended use. After approval from the IRB (ethical committee), the study was conducted under the supervision of Prof. & Head of the department obstetrics and gynecology GOVERNMENT MEDICAL COLLEGE, AKOLA.

The survey proforma was prepared to collect the data relevant to the study. No confidential data was collected. The willing participants after human informed consent were enrolled in the study. The sterile (Autoclaved) PPH cup was inserted in the birth canal (vagina) by paramedics or obstetricians to measure the post delivery blood loss. The cup was kept in situ for to 2 to 12 hours as per the institution protocol for intensive monitoring for PPH depending on the type of the delivery.

Risk assessment :

Discomfort if any was minimized by using sterile xylocaine jelly to lubricate the device .The risk of infection was minimized by using sterile PPH cup; it was autoclaved along with the medical instruments to be used before every use.

Benefits of the study :

1) In case of excessive blood loss, due to early diagnosis the adequate medical measures were taken so future morbidity and mortality was minimized.
2) The post delivery blood loss was measured in real time and accurately. It had an advantage over the prevalent methods.
3) There was ease in nursing care of the patients as no need of sanitary pads and handling was minimized.
4) The participants had dry feel post delivery and no soiling of cloths due to blood.

Protection of privacy :

As per the attached survey proforma the data was collected related to the acceptance and user friendliness of the PPH cup to the patients, paramedics and obstetricians. No confidential information was collected. Only duly filled survey proforma was communicated to the student for further analysis.

Informed consent process :

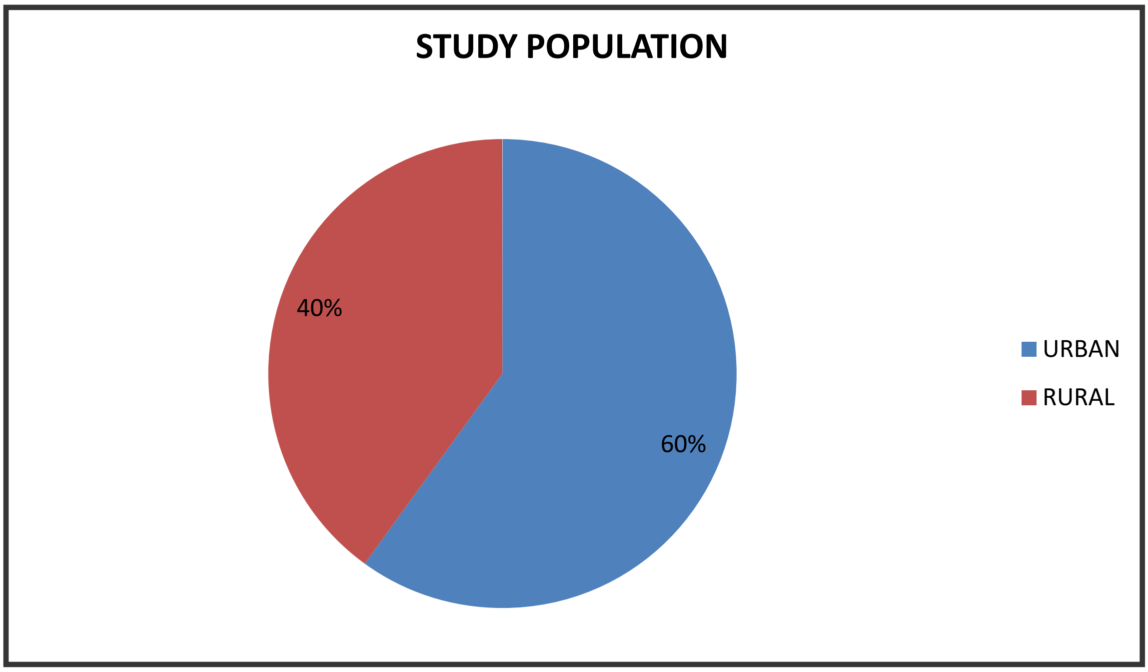
The participants were given information about the study using the human informed consent form. If any queries, they were provided the contact details of the Prof. & HOD of department of obstetrics and gynecology, Govt. Medical College, Akola. They were told that the sterile PPH cup will be inserted in the birth canal (vagina) by the paramedics or obstetricians to measure the blood loss after childbirth to monitor the blood loss. It will be kept for to 2 to12 hours. The benefits & measures to minimize the possible risk were explained in local language so that they understand it. They were also told that their participation is voluntary for enrollment in the study. All queries were resolved. They were also explained that they have a right to stop the participation at any time and will not affect their treatment. If the participant was willing to voluntarily participate in the study the prescribed human consent form was signed and enrolled in the study. No confidential information was collected and not communicated to anyone.

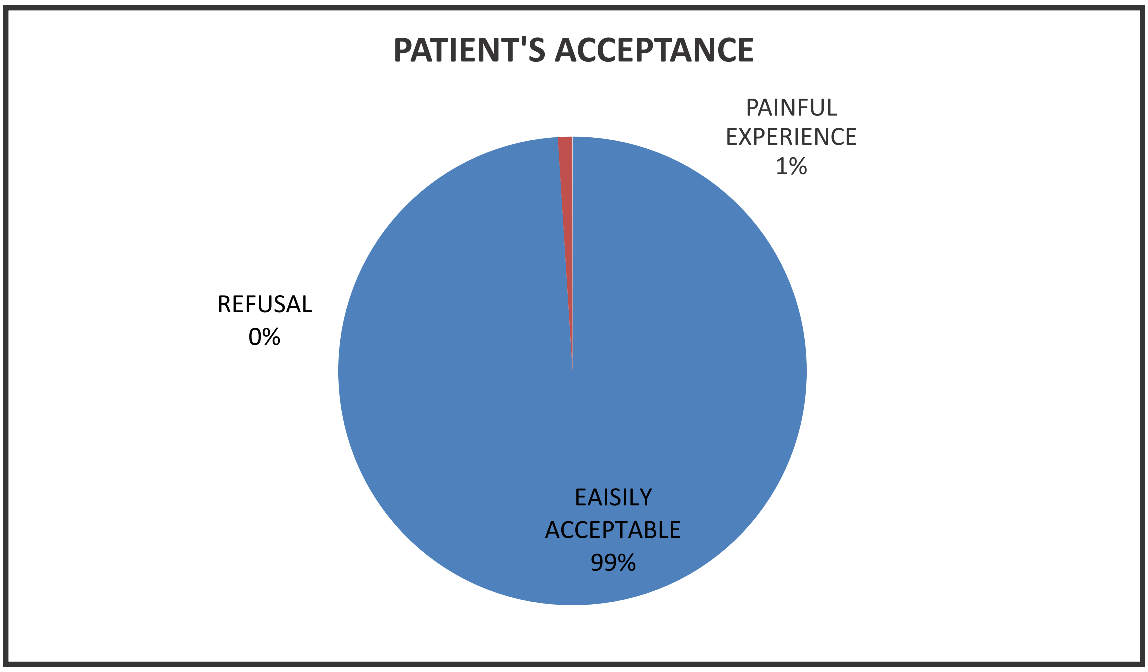
Data collection and analysis :

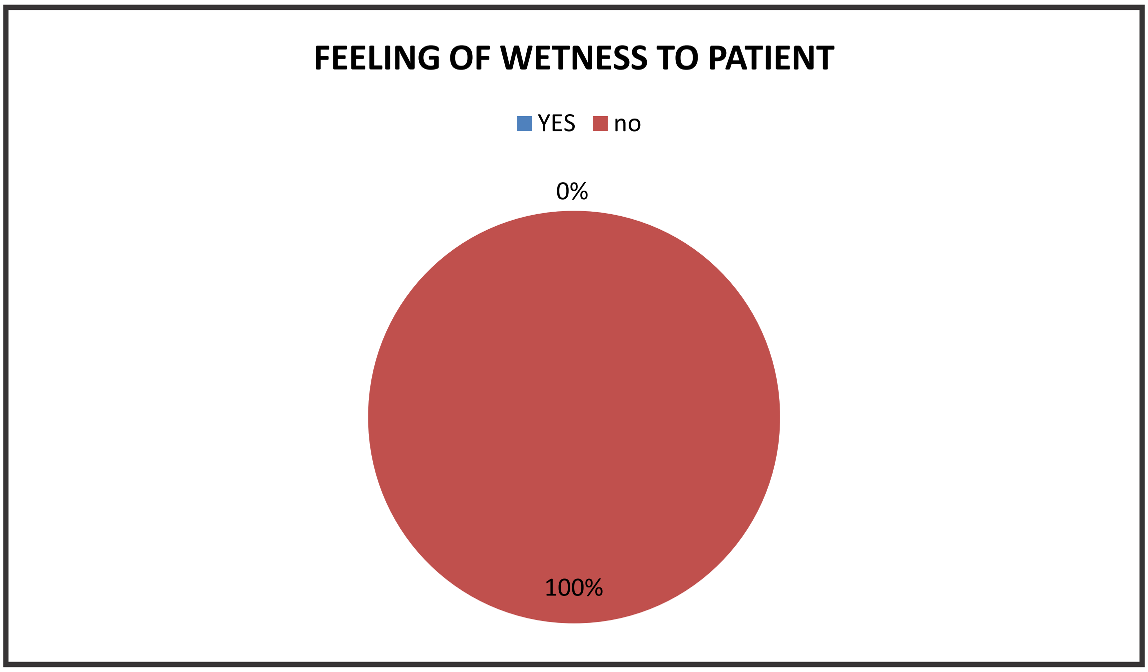
The manually collected data from the survey proforma was transferred to excel sheet and later analyzed to get tables and charts.

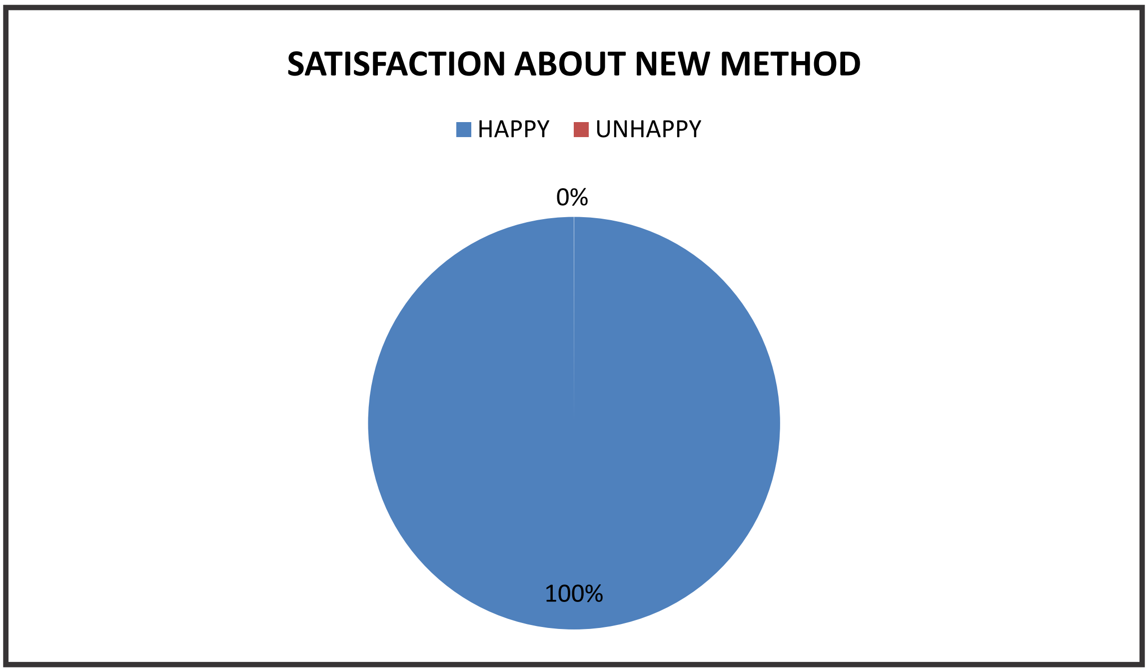
Observations :

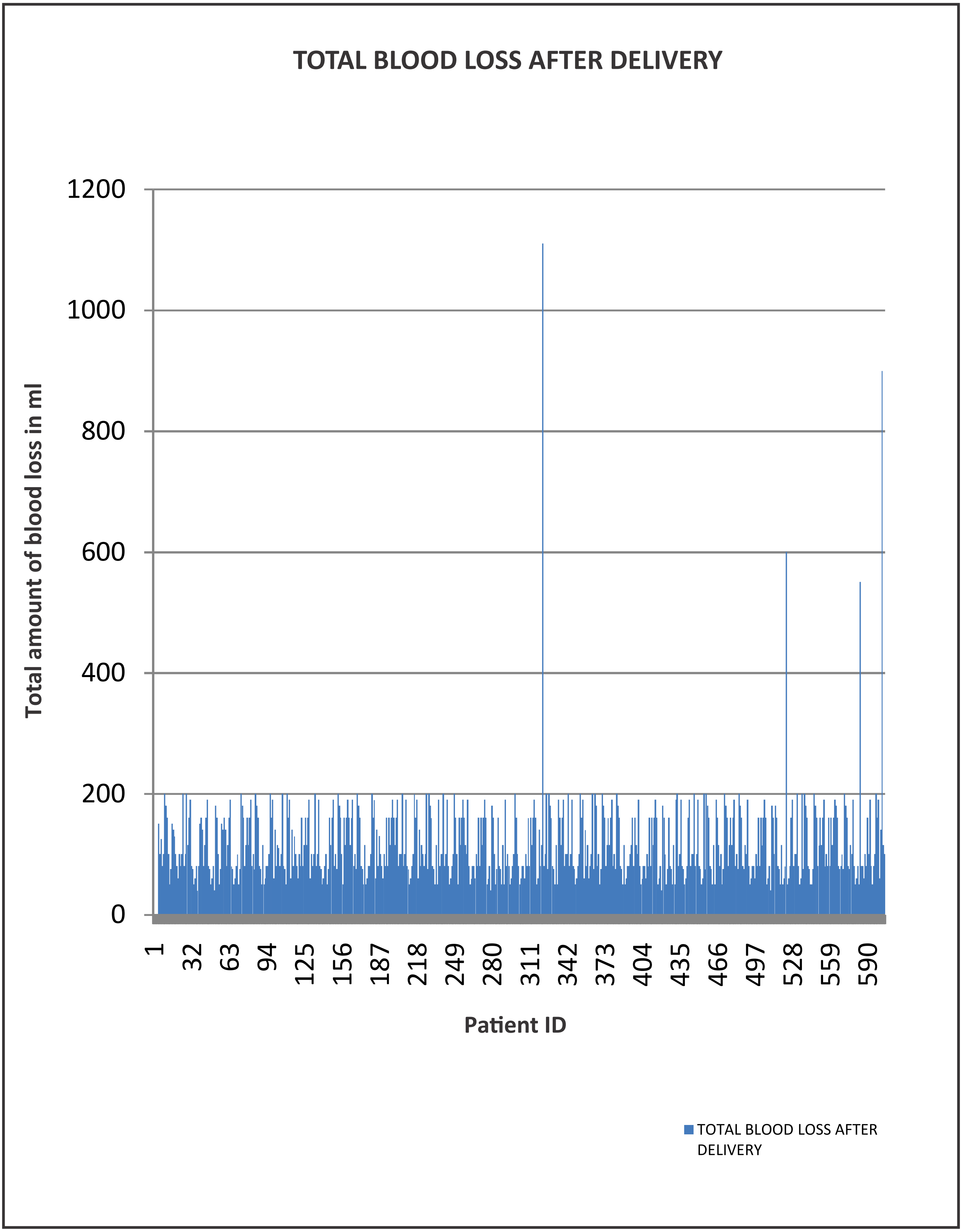
PATIENT RELATED PARAMETERS





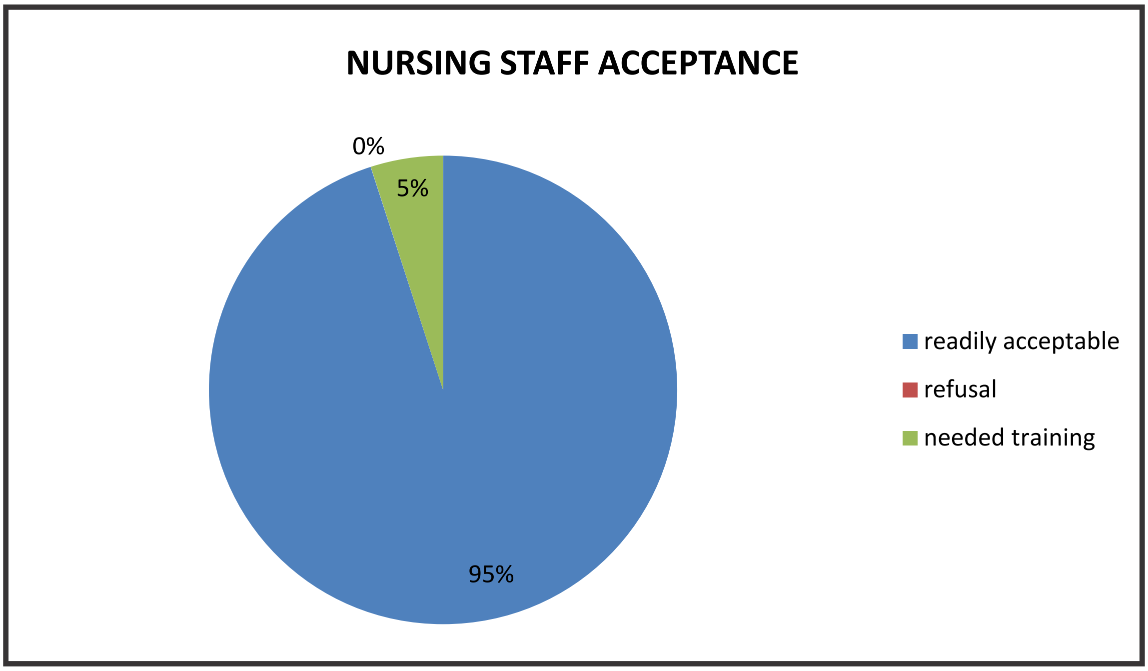

FEELING OF WETNESS TO PATIENT
YES
no
0%
100%

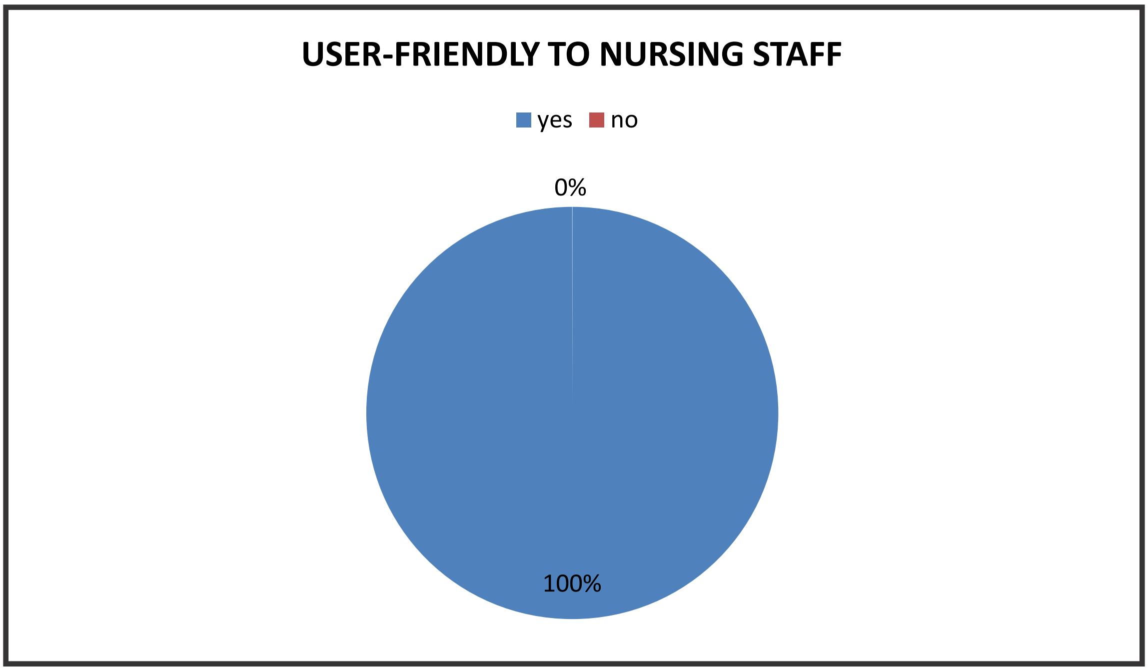

SATISFACTION ABOUT NEW METHOD
HAPPY
UNHAPPY
0%
100%

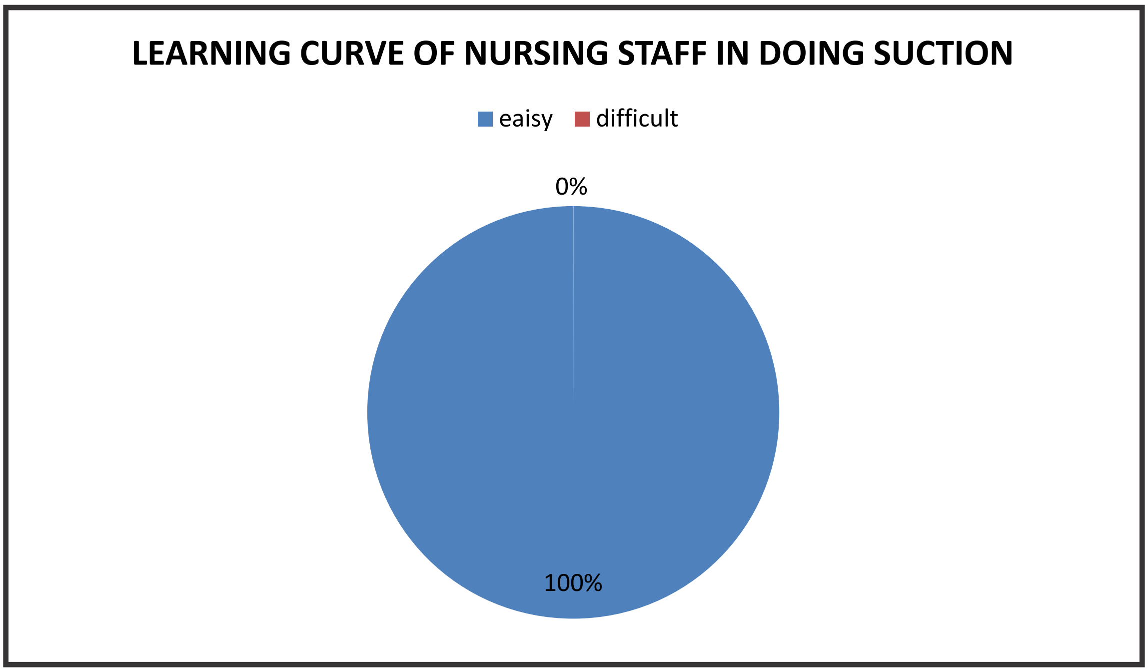


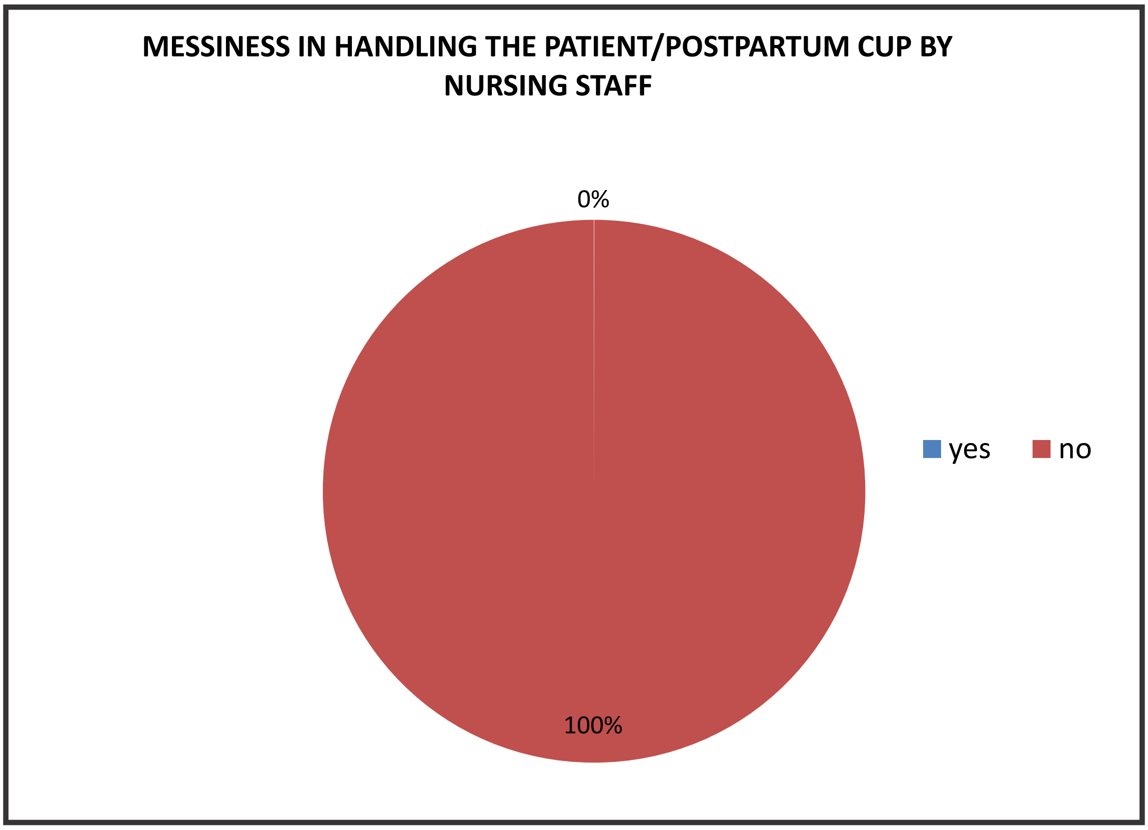
TOTAL BLOOD LOSS AFTER DELIVERY
1200
1000
800
600
400
200
0
Total amount of blood loss in ml
1
32
63
94
125
156
187
218
249
280
311
342
373
404
435
466
497
528
559
590
Patient ID
TOTAL BLOOD LOSS AFTER DELIVERY

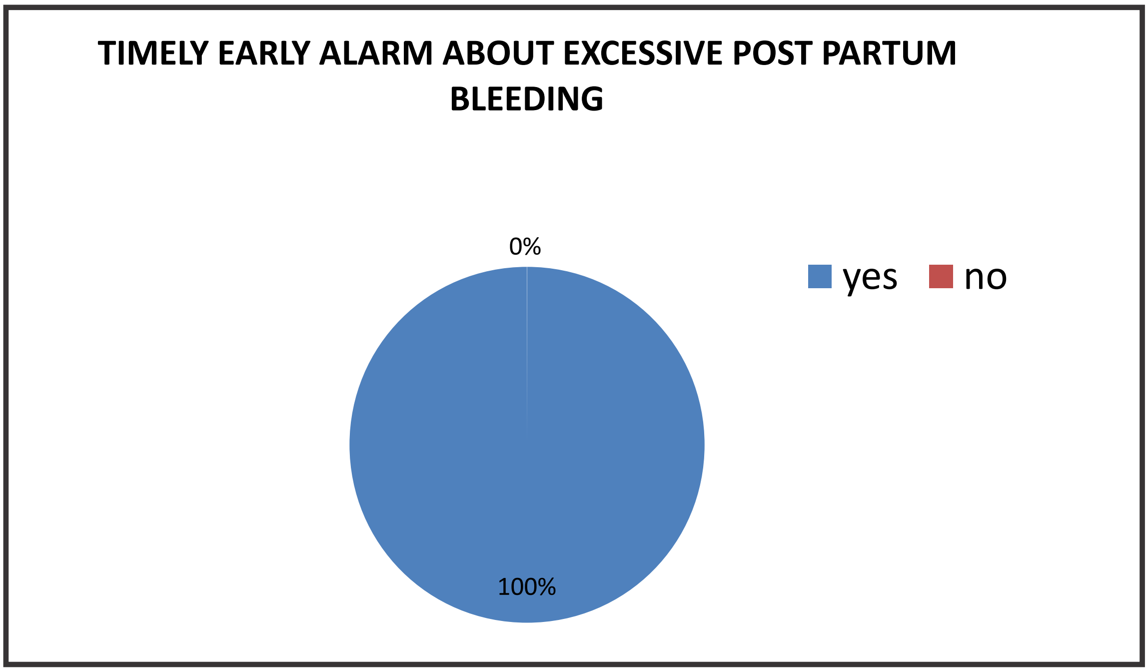
NURSING STAFF RELATED PARAMETERS

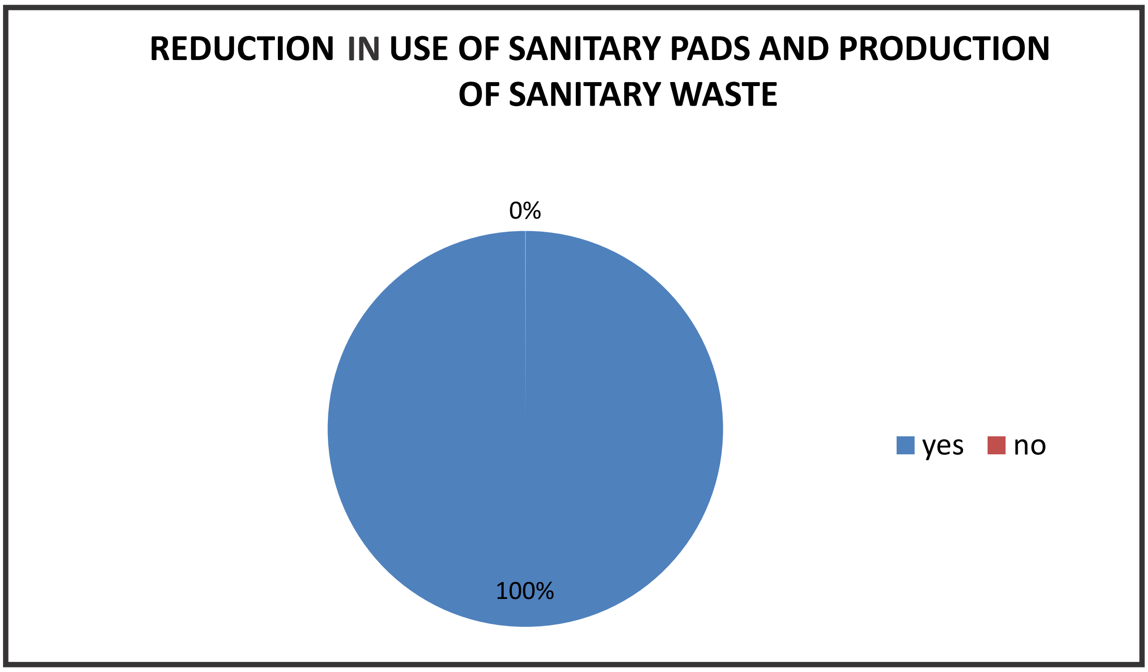




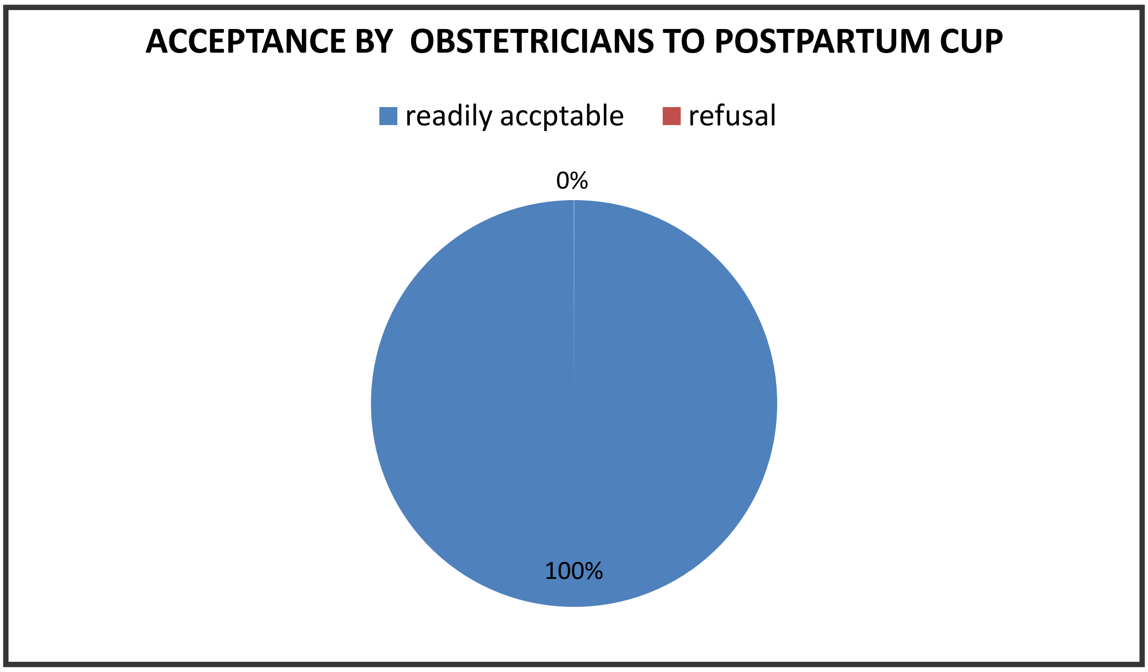

LEARNING CURVE OF NURSING STAFF IN DOING SUCTION
eaisy
difficult
0%
100%

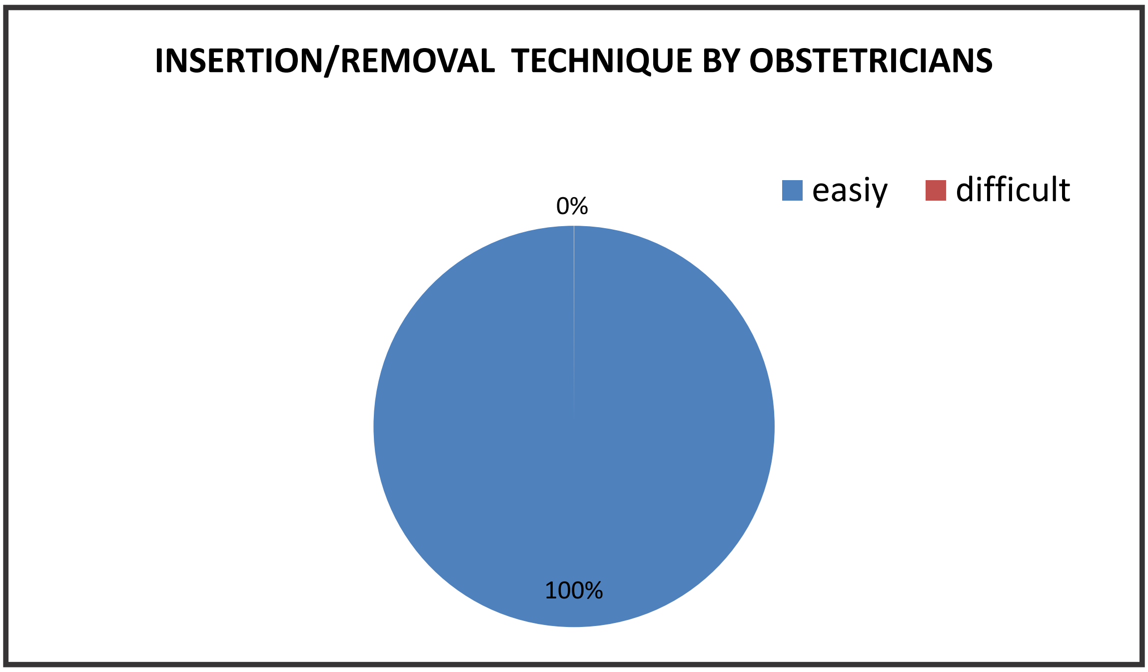

MESSINESS IN HANDLING THE PATIENT/POSTPARTUM CUP BY NURSING STAFF
0%
100%
yes
no

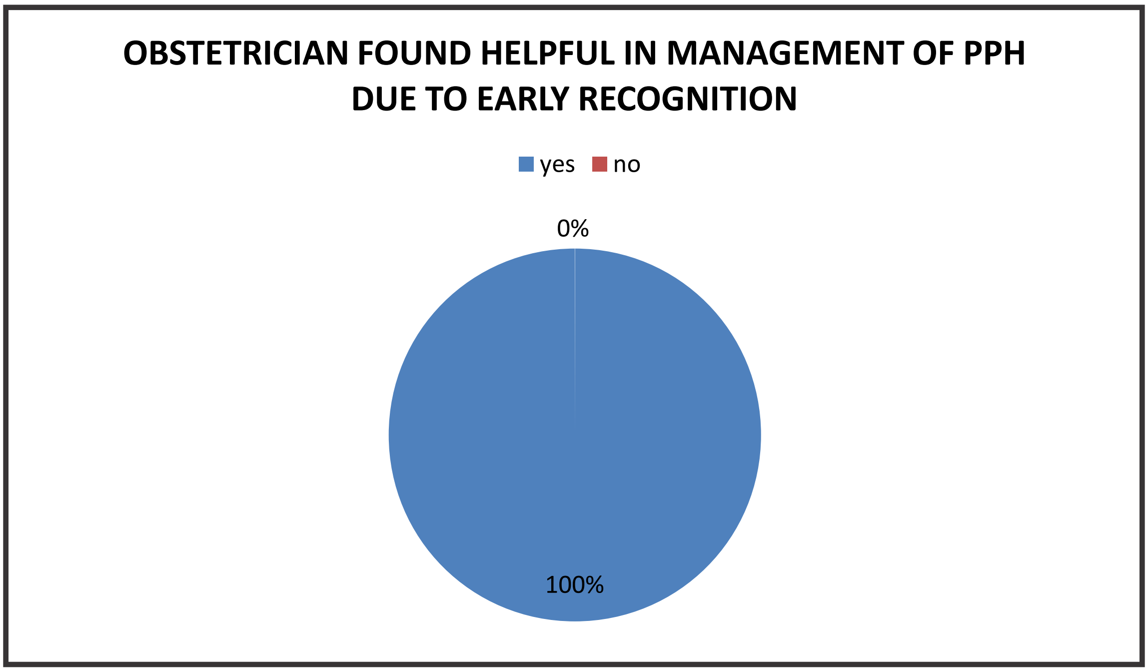

TIMELY EARLY ALARM ABOUT EXCESSIVE POST PARTUM BLEEDING
0%
yes
no
100%

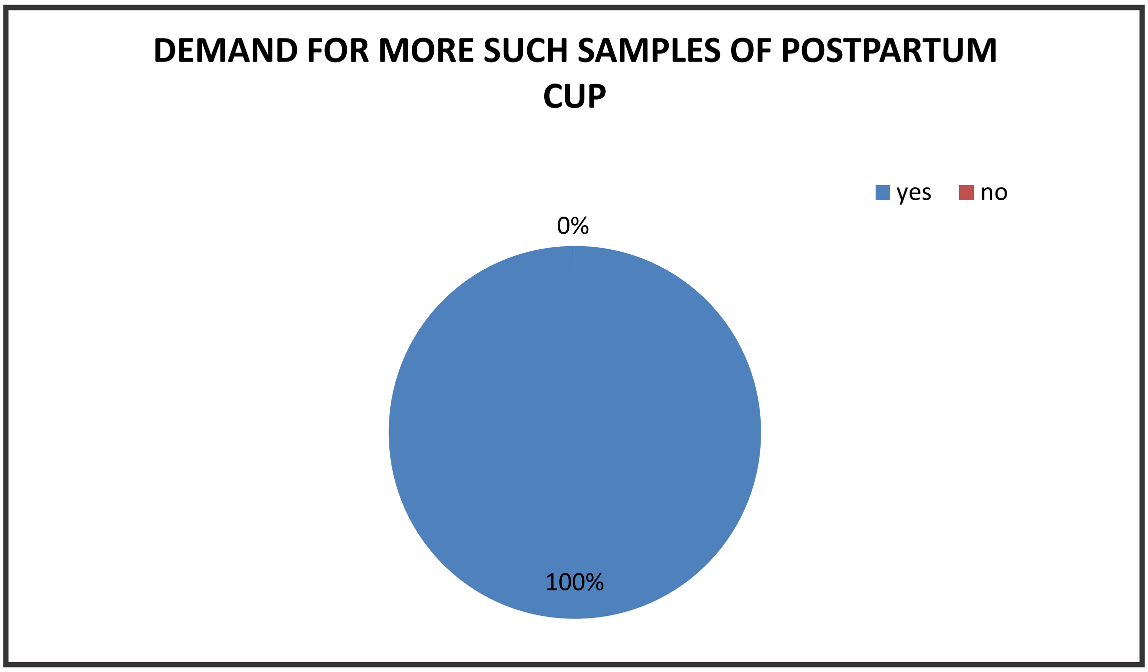

REDUCTION IN USE OF SANITARY PADS AND PRODUCTION OF SANITARY WASTE
0%
yes
no
100%

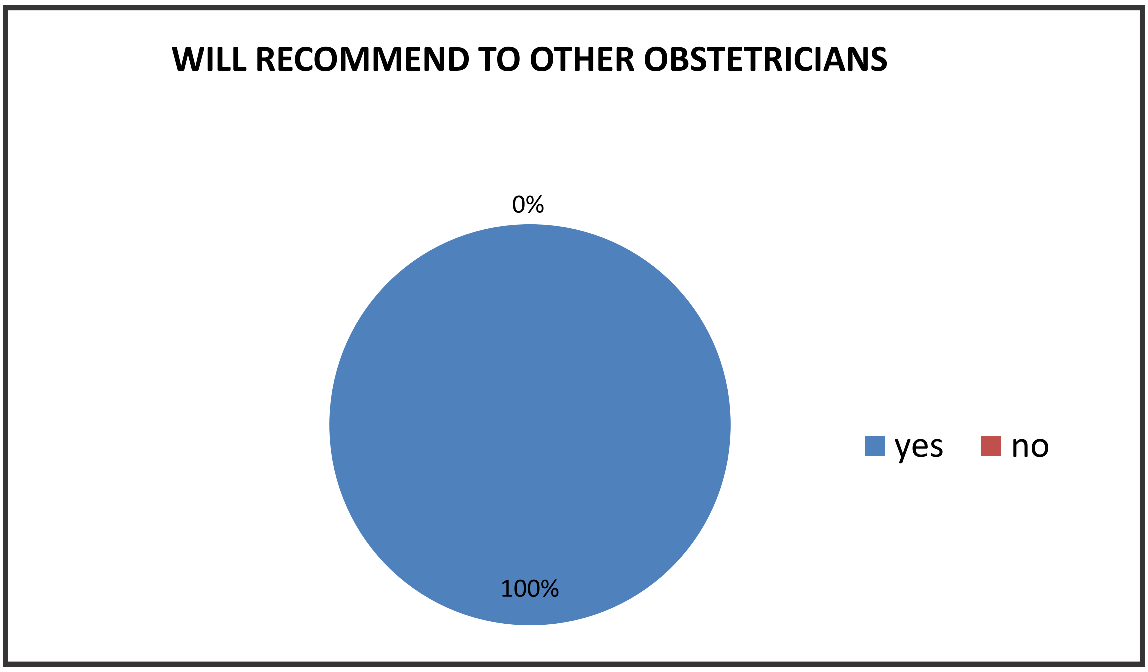
OBSTETRICIAN RELATED PARAMETERS

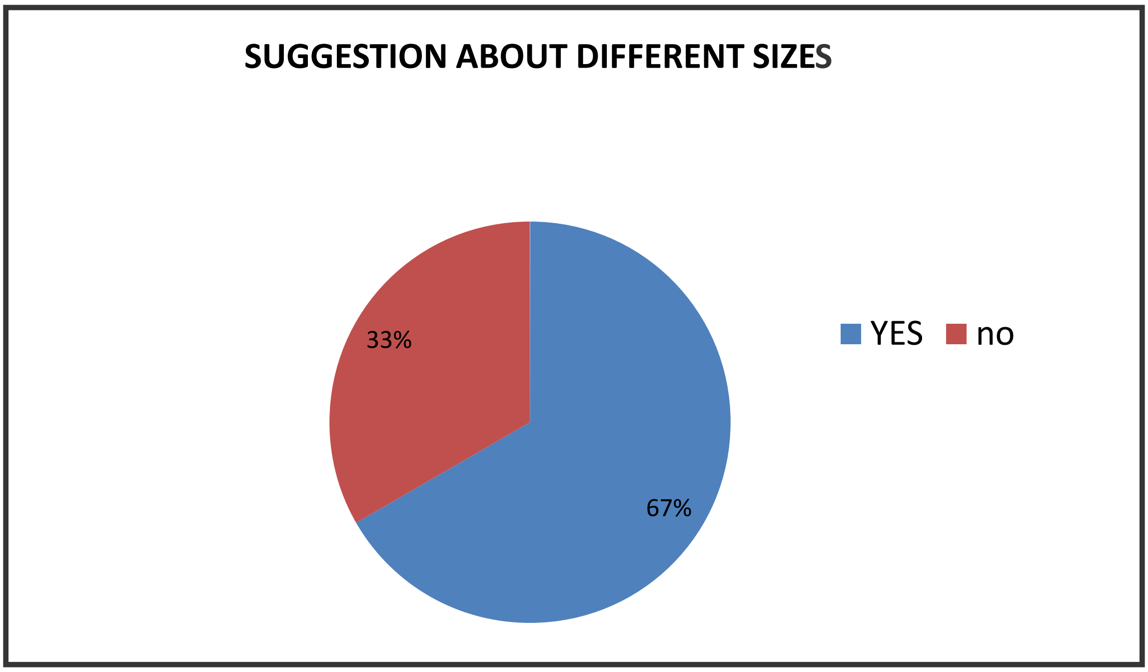


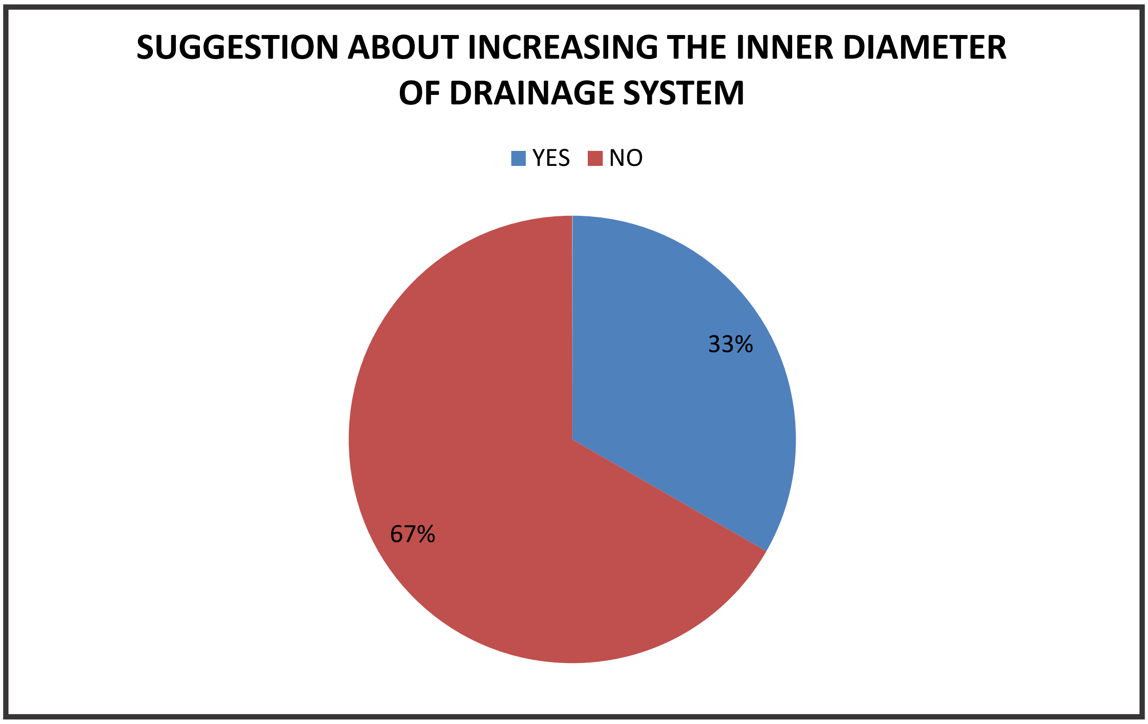


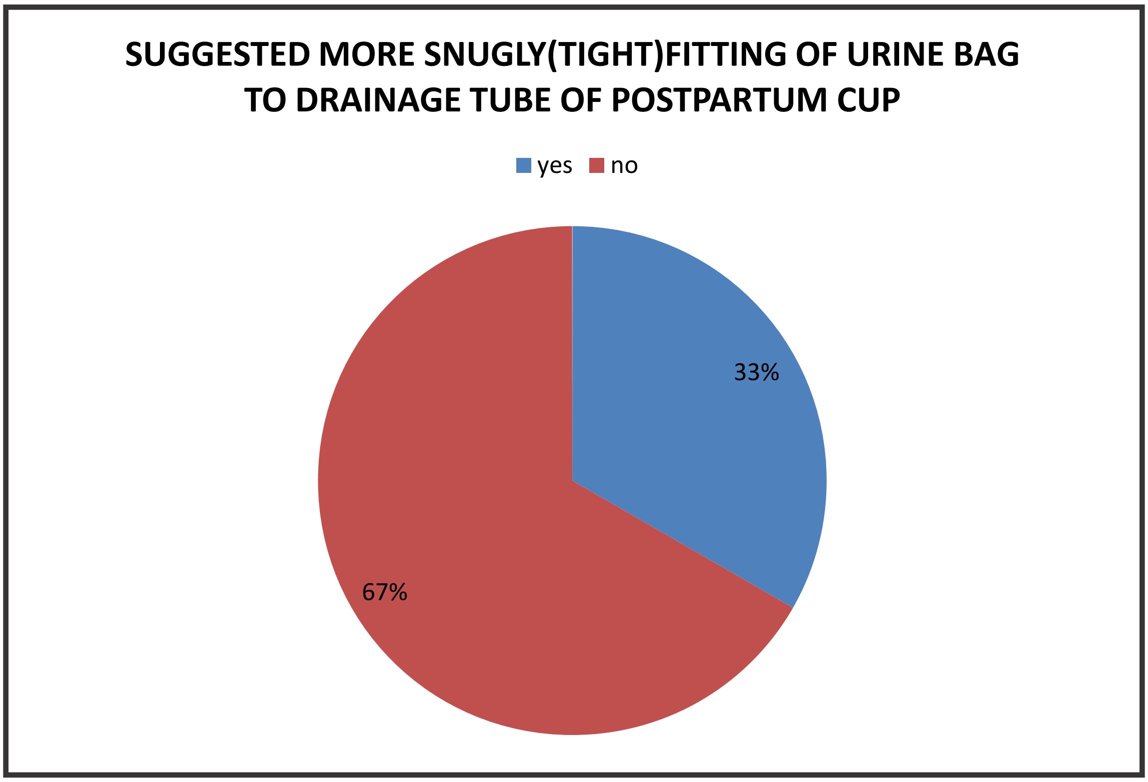

OBSTETRICIAN FOUND HELPFUL IN MANAGEMENT OF PPH
DUE TO EARLY RECOGNITION
yes
no
0%
100%


DEMAND FOR MORE SUCH SAMPLES OF POSTPARTUM
CUP
yes
no
0%
100%

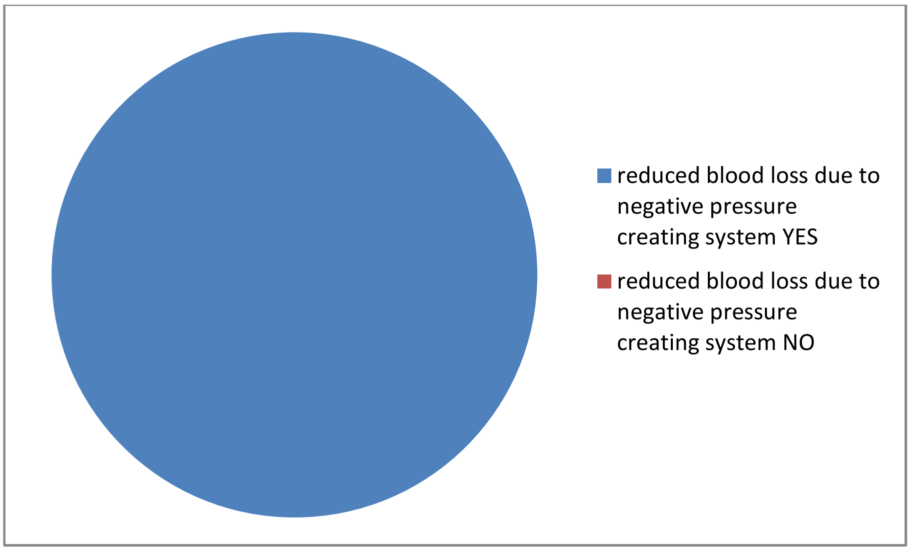

WILL RECOMMEND TO OTHER OBSTETRICIANS
0%
100%
yes
no


SUGGESTION ABOUT DIFFERENT SIZES
33%
67%
YES
no


SUGGESTION ABOUT INCREASING THE INNER DIAMETER OF DRAINAGE SYSTEM
YES
NO
33%
67%


SUGGESTED MORE SNUGLY(TIGHT)FITTING OF URINE BAG TO DRAINAGE TUBE OF POSTPARTUM CUP
yes
no
33%
67%

Reduced blood loss due to negative pressure



CONCLUSION

After data analysis I concluded,

ADVANTAGES :

1. The postpartum cup was readily acceptable to patients, user friendly to nursing staff and obstetricians.
2. Patients were extremely happy after its use as there was total dry feel after delivery. Very less handling of patients while checking the obstetric blood loss.
3. It was safe and effective in real-time estimation of post partum bleeding which was not possible with prevalent methods.
4. The obstetricians were alarmed early about excessive post partum bleeding so patients were treated without losing golden hour.
5. There was reduction in use of sanitary pads so less sanitary waste was generated.
6. Obstetricians easily accepted this novel idea to be implemented in clinical practice, they found insertion and removal techniques very easy. The cup formed watertight seal with vaginal walls and there was absolutely no leakage of blood from the side of postpartum cup.
7. Through suction port, easy suction was possible; so no blockage of the system due to blood clots. Especially more useful in overweight patients as there was no handling of patient.
8. The obstetricians, who conducted the trial were happy using it and demanded more such postpartum cup. They also recommended use of postpartum cup to their colleagues.
9. Suggested to be useful in Low-Resource and primary health care settings.

LIMITATIONS / SUGGESTIONS :

Although there were no limitations, however few suggestions such as,

1) Different sizes should be made available.
2) The inner diameter of the suction port should be such that 10 mm suction tube should negotiate easily the joint of drainage end and suction end of the postpartum cup.
3) The collection bag should be tightly attached to drainage end of the cup to make it water tight.
4) In case of vaginal or perineal tears, the blood loss will not be collected into the cup. It will soil the perineum with blood. So in any such cases the trauma should always be ruled out.

Above mentioned suggestions were implemented by modifications in the designs of same mould.

More sizes of the cup were made available.

Future application :

The result of the pilot study needs to be confirmed by randomised control multi centre trials. The further studies will prove it's effectiveness in measuring the blood loss and ease of use to the women and obstetricians.

The observation by the experts that the use of the PPH cup has reduced the post delivery blood loss by the virtue of its ability to act as negative pressure creating system to suck out the blood clots retained inside the uterine cavity needs to be verified by further clinical trials.

REFERENCES :

1. Centres for Disease Control and Prevention. Reproductive Health: Pregnancy Mortality Surveillance System.
Available at https://www.cdc.gov/reproductivehealth/maternalinfanthealth/pmss.html.
June 29, 2017; Accessed: July 21, 2017.

2. WHO. Reducing the Global Burden: Postpartum Haemorrhage. Making Pregnancy Safer. 2007. [Full Text].

3. American College of Obstetricians and Gynaecologists.
ACOG Practice Bulletin: Clinical Management Guidelines for Obstetrician-Gynaecologists Number 76, October 2006: postpartum hemorrhage. Obstet Gynecol. 2006 Oct. 108(4):1039-47. [Medline].

4. Lutomski J, Byrne B, Devane D, Greene R. Increasing trends in atonic postpartum haemorrhage in Ireland: an 11-year population-based cohort study. BJOG. 2012 Feb. 119(3):306-14. [Medline].

5. Baskett TF. Complications of the third stage of labour. Essential Management of Obstetrical Emergencies. 3rd ed. Bristol, England: Clinical Press; 1999. 196-201.

6. Sentilhes L, Vayssière C, Deneux-Tharaux C, et al. Postpartum hemorrhage: guidelines for clinical practice from the French College of Gynaecologists and Obstetricians (CNGOF): in collaboration with the French Society of Anesthesiology and Intensive Care (SFAR). Eur J Obstet Gynecol Reprod Biol. 2016 Mar. 198:12-21. [Medline].

7) Diaz_V, Abalos_E, Carroli_G. Methods for blood loss estimation after vaginal birth. Cochrane Database of Systematic Reviews 2018, Issue 9. Art.No.: CD010980. DOI:10.1002/14651858.CD010980.pub2

SURVEY OF PPH CUP USE

ID / HOSPITAL REGISTRATION NO : ☐

A) PATIENT RELATED PARAMETERS

1) Urban ☐ Rural ☐ Age : ☐ yrs.

2) ACCEPTANCE

Easily Acceptable ☐ Discomfort ☐ Refusal ☐

3) WETNESS

FEELING OF WETNESS TO PATIENTS

Yes ☐ No ☐

B) NURSING STAFF RELATED PARAMETERS

1) ACCEPTANCE / TRAINING

Readily Acceptable ☐ Need Training ☐ Refusal ☐

2) USER FRIENDLY TO NURSING STAFF

Yes ☐ No ☐

3) LEARNING CURVE OF NURSING STAFF IN DOING SUCTION

Easy ☐ Difficult ☐

4) MESSINESS IN HANDLING THE PATIENT / PPH CUP BY NURSING STAFF

Yes ☐ No ☐

5) TIMELY EARLY ALARM ABOUT EXCESSIVE POSTPARTUM BLEEDING

Yes ☐ No ☐

6) REDUCTION IN USE OF SANITARY PADS AND GENERATION OF BIOMEDICAL WASTE

Yes ☐ No ☐

C) OBSTETRICIAN RELATED PARAMETERS

1) ACCEPTANCE BY OBSTETRICIANS TO PPH CUP

Yes ☐ No ☐

2) INSERTION / REMOVAL TECHNIQUE BY OBSTETRICIAN

Easy ☐ Difficult ☐

3) DEMAND FOR MORE SUCH SAMPLE

Yes ☐ No ☐

4) WILL RECOMMEND TO OTHER

Yes ☐ No ☐

5) SUGGESTIONS ABOUT DIFFERENT SIZES

Yes ☐ No ☐

6) SUGGESTIONS ABOUT INCREASING INNER DIAMETER OF THE DRAINAGE SYSTEM

Yes ☐ No ☐

7) SUGGESTED MORE SNUGLY FITTING COLLECTION BAG

Yes ☐ No ☐

E) TOTAL BLOOD LOSS AFTER DELIVERY

☐ ML

D) REMARKS BY OBSTETRICIANS

1) USEFUL AT PRIMARY AND LOW RESOURCE SETTINGS.

2) EARLY ALARM & PROMPT TREATMENT REDUCE NEED FOR THE BLOOD TRANSFUSION.

3) REDUCE BLOOD LOSS AFTER DELIVERY.

4) MATERNAL MORBIDITY / MORTALITY REDUCED.